

Literature Status for arXiv:1207.0975, Question 4.1

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Status. This is a literature-already-answered packet. The answer is known in a later paper and is not claimed as a new proof from this run.

Source question. Fritz, Netzer, and Thom ask in arXiv:1207.0975, Question 4.1, whether for

$$\Gamma = \mathbb{F}_2 \times \mathbb{F}_2$$

the universal group-C*-norm function

$$\mathbb{Z}\Gamma \ni a \mapsto \|a\|_u \in \mathbb{R}$$

is computable.

Supporting answer. Goldbring and Sinclair prove in arXiv:2509.15086, Corollary **FNTquestion**, that for every $n \geq 2$, including $n = \infty$, the standard presentation of $C^*(\mathbb{F}_n \times \mathbb{F}_n)$ is not computable. They then state explicitly that the case $n = 2$ answers the question posed by Fritz, Netzer, and Thom.

Identification. For the standard generators of $\mathbb{F}_2 \times \mathbb{F}_2$, generated points in the standard presentation of $C^*(\mathbb{F}_2 \times \mathbb{F}_2)$ are precisely finite rational *-polynomials, equivalently rational group-ring elements, in those generators. A computable standard presentation would therefore give an algorithm that approximates $\|a\|_u$ for each such a to arbitrary precision. Conversely, the source question's computability of $a \mapsto \|a\|_u$ is exactly the norm-computability part of that standard presentation. Thus Goldbring–Sinclair's corollary gives a negative answer to Question 4.1.

Scope. This packet concerns the $F_2 \times F_2$ computability question in Section 4 of arXiv:1207.0975. It does not address the separate question in the same paper about $SL_3(\mathbb{Z})$.

References

- [1] T. Fritz, T. Netzer, and A. Thom, *Can you compute the operator norm?*, Proc. Amer. Math. Soc. 142 (2014), 4265–4276; arXiv:1207.0975.
- [2] I. Goldbring and T. Sinclair, *On definability of C*-tensor norms*, arXiv:2509.15086.